# Basic Mathematics Ep. 1 — Natural Numbers & Integers

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## Core concepts

Natural numbers came from the need to count; integers came from the need to subtract freely.

- **Natural numbers:** N = {1, 2, 3, 4, ...}. Early humans used tally marks, then invented numbers to count tribe members and food.
- **Integers:** Z = {..., -3, -2, -1, 0, 1, 2, 3, ...}. With only N, 10 - 12 had no answer. Debt gave negative numbers; "nothing left" (5 - 5) gave zero.
- **No largest natural number (proof by contradiction):** assume n is the largest. Then n + 1 is also natural and bigger, a contradiction.
- **Method of contradiction, everyday example:** someone says the water is very hot, yet you can dip a finger in comfortably. The second fact contradicts the first.
- **Set notation:** $\in$ means "belongs to", $\notin$ means "does not belong to". $2 \in N$; $-3 \notin N$ but $-3 \in Z$; $-1/2 \notin Z$.
- **Axioms:** at this level, the properties below are accepted as self-evident. They matter because some fail later (order property for complex numbers; some vector products).
- India is credited with the decimal system.

## Properties of integers

These laws hold for natural numbers, integers and real numbers; each can be "felt" with balls, line segments or areas.

| Property | Rule | How to feel it |
|---|---|---|
| Commutative (addition) | a + b = b + a | Join segments a and b in either order; same length |
| Additive identity | a + 0 = 0 + a = a | Adding nothing to 3 balls leaves 3 |

| Property | Rule | How to feel it |
|---|---|---|
| Associative (addition) | (a + b) + c = a + (b + c) | Group segments either way; same total |
| Distributive | a(b + c) = ab + ac | Rectangle a by (b + c), split into two parts |
| Commutative (multiplication) | ab = ba | Rotate an a by b rectangle; same area |
| Multiplicative identity | a × 1 = 1 × a = a | Add 1 a times, or add a once |
| Zero product | a × 0 = 0 × a = 0 | Add 0 a times, or add a zero times |
| Associative (multiplication) | (ab)c = a(bc) | Count cubes in a box two ways |

**Four operations:** addition is the most basic; subtraction is its reverse; multiplication is repeated addition (ab = a added b times); division is the reverse of multiplication.

**Distributive example:** 2 × (3 + 4) = 2 × 7 = 14, and 2×3 + 2×4 = 6 + 8 = 14.

### Associative property of multiplication: (ab)c = a(bc)

The grouping does not change the product. Example: (2 × 3) × 4 = 6 × 4 = 24, and 2 × (3 × 4) = 2 × 12 = 24.

Picture a box of small cubes, 2 long, 3 wide and 4 high:

1. **Layer by layer:** one floor has 2 × 3 = 6 cubes; 4 layers give (2 × 3) × 4 = 24.
2. **Slice by slice:** one front slice has 3 × 4 = 12 cubes; 2 slices give 2 × (3 × 4) = 24.

Same box, same count. So a × b × c is the volume of a cuboid, seen two ways.

**Use it to calculate faster:** 25 × 17 × 4 = 17 × (25 × 4) = 1700.

**Reference video:** *How the Associative Property Works* (about 1 min 23 s). It animates exactly this 2 × 3 × 4 block being sliced two ways. Watch it alongside this section.

## Sign rules

Positive × negative is negative, and negative × negative is positive; both follow from the properties above.

**Positive × negative = negative.** 3 × (-4) means adding -4 three times: -4 - 4 - 4 = -12. In general a × (-b) = -ab. Think of borrowing ₹b, a times over.

**Proof that (-a)(-b) = ab:**

1. Anything times 0 is 0: (-a) × 0 = 0.
2. Write 0 as b + (-b): (-a) × [b + (-b)] = 0.
3. Distributive law: (-a)(b) + (-a)(-b) = 0.
4. Since (-a)(b) = -ab: -ab + (-a)(-b) = 0.
5. Add ab to both sides: (-a)(-b) = ab.

**"Changing sides changes the sign"** is only a shortcut. What actually happens is adding or subtracting the same quantity on both sides, which keeps the equation balanced.

**Exponent facts (used in Problem 8):**

- $a^0 = 1$ for any $a \neq 0$, because $a^0 = a^1/a^1$.
- A positive base raised to any power stays positive.
- $2^x < 1$ when x is negative ($2^{-1} = 1/2$, $2^{-2} = 1/4$).

## Solved problems

Each problem uses only natural numbers and integers, and most are a modified version of the one before it.

| # | Problem | Key step | Answer |
|---|---|---|---|
| 1 | xy = x + y (integers) | Add 1: (x - 1)(y - 1) = 1; both factors 1 or both -1 | (2, 2), (0, 0) |
| 2 | xy - x + y = 3 (integers) | Subtract 1: (x + 1)(y - 1) = 2; pairs ±(1, 2), ±(2, 1) | (1, 2), (0, 3), (-3, 0), (-2, -1) |
| 3 | xy + 2x - 2y = 2031 (integers) | Subtract 4: (x - 2)(y + 2) = 2027, which is prime | (2029, -1), (3, 2025), (-2025, -3), (1, -2029) |
| 4 | $x^2 + y^2 + xy(x + y + 2) = 6$ (naturals) | (x + y)(x + y + xy) = 6; only 2 × 3 works, so xy = 1 | (1, 1) |
| 5 | xy + 1 = x + y (integers) | (x - 1)(y - 1) = 0, so one factor is 0 and the other is free | x = 1 with any y, or y = 1 with any x |

| # | Problem | Key step | Answer |
|---|---|---|---|
| 6 | $5xy + 3z = 14$ (naturals) | Try $z = 1$ to 4; only $z = 3$ gives $xy = 1$ | (1, 1, 3) |
| 7 | $2xy + 4z = z(x^2 + 4y^2)$ (naturals) | $z \geq 1$ gives $(x - y)^2 + 3y^2 \leq 4$, so $y = 1$, $x = 1$ or 2 | (1, 1, 2), (2, 1, 1) |
| 8 | $2^x + 1 = y^2$ (integers) | $x \leq 0$ fails; for $x > 0$, $y - 1$ and $y + 1$ are powers of 2 differing by 2 | (3, 3), (3, -3) |
| 9 | Show $x^3 = y^2 + z^4$ has natural solutions | $2^9 = 2^8 + 2^8$, so $8^3 = 16^2 + 4^4$ | $x = 8$, $y = 16$, $z = 4$ |
| 10 | $(ab + bc + ca)/(a + b + c)$ integer $\Rightarrow$ $(a^2 + b^2 + c^2)/(a + b + c)$ integer | Divide $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ by $(a + b + c)$ | Integer minus 2 × integer |
| 11 | $(3a + 4b)/5$ integer $\Rightarrow$ $(4a - 3b)/5$ integer | $3 \times (3a + 4b)/5 - (a + 3b) = (4a - 3b)/5$ | Integer |

**Notes on tricky steps:**

- **Problem 5 is a trap.** Most students answer only (1, 1). If $ab = 0$, either factor can be 0 while the other is any integer, including 0.
- **Problem 7:** $z = 2xy/(x^2 + 4y^2 - 4)$. The denominator is at least 1, so $z \geq 1$ gives $x^2 + 4y^2 - 4 \leq 2xy$. $x = 0$ is rejected because it is not natural.
- **Problem 8:** $(y - 1)$ cannot be 1, because then $2^x = 3$, and even ≠ odd. From $2^p(2^{(q-p)} - 1) = 2$, the bracket is odd, so it must be 1: $p = 1$, $q = 2$, giving $y = 3$ and $x = 3$.
- **Problem 9 generalized:** $x^n = y^{n-1} + z^{n+1}$ is solved by $x = 2^n$, $y = 2^{n+1}$, $z = 2^{n-1}$, using $n^2 - 1 = (n - 1)(n + 1)$.
- **Problem 10:** assume $a + b + c \neq 0$ whenever it is a denominator.

## Strategies and homework

Seven tricks solve every problem in this lecture.

1. **Add or subtract a constant to factorize** into (integer) × (integer) = number.
2. **Prefer a prime on the right side:** few factor pairs. Always include negative pairs for integers.

3. **ab = 0** means a = 0 (b anything) or b = 0 (a anything).
4. **Bound the variables** for natural numbers: positive terms cannot exceed the total.
5. **Squares are never negative;** use this to trap values in an inequality.
6. **Parity:** an even number never equals an odd one.
7. **Check every case** systematically so no root is lost.

**Must-remember identity:** $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.

**Homework**

- [ ] Show (ab)c = a(bc) using the volume of a cuboid.
- [ ] Explain why (-) × (+) is negative.
- [ ] Is this proof valid? $(-1)(-1) = \cos \pi \cdot \cos \pi = [\cos 2\pi + \cos 0]/2 = 1$. Does the cosine formula itself depend on the sign rule?
- [ ] Find other ways to build (4a - 3b)/5 from (3a + 4b)/5, or show this is the only one.
- [ ] Do the episode assignment: superinfinity.com/assignment/1

**Next episode:** even and odd integers.